

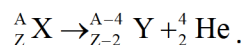
Chapter Three

Radioactivity

Experiments performed subsequently showed that radioactivity was a nuclear phenomenon in which an unstable nucleus undergoes a decay. This is referred to as radioactive decay. Three types of radioactive decay occur in nature:

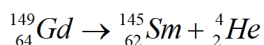
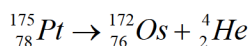
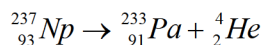
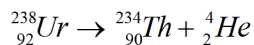
(i) Alpha Decay

α -decay in which a helium nucleus ${}^4_2\text{He}$ is emitted;



Where ${}^A_Z\text{X}$ parents' nucleus, ${}^{A-4}_{Z-2}\text{Y}$ is the daughter and ${}^4_2\text{He}$

A is total no of Nucleons, Z is the total no of proton.



(ii) Beta decay

In beta decay, a nucleus spontaneously emits an electron (β^- decay) or a positron (β^+ decay).

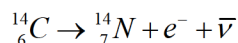
(1) Beta-minus decay

In beta minus, a neutron is transformed into proton causing increase of atomic number. The change in atomic configuration is given by ${}_Z^AX \rightarrow {}_{Z+1}^AY + e^- + \bar{\nu}$

To maintain the conservation of charge the nucleus process also an electron and antineutrino

$$N = P + e^- + \bar{\nu}$$

A common example of β^- decay is ${}_{15}^{32}P \rightarrow {}_{16}^{32}S + e^- + \bar{\nu}$



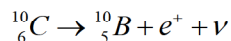
(2) Beta-plus decay

In beta plus, a proton is transformed into neutron causing decrease of atomic number. The change in atomic configuration is given by ${}_Z^AX \rightarrow {}_{Z-1}^AY + e^+ + \nu$

To maintain the conservation of charge the nucleus process also an electron and antineutrino.

$$P = N + e^+ + \nu$$

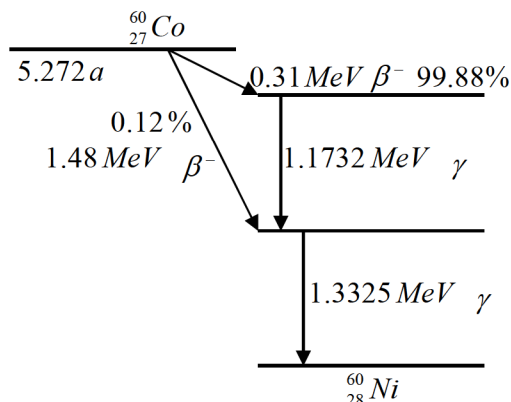
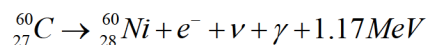
A common example of β^+ decay is ${}_{11}^{22}Na \rightarrow {}_{10}^{22}Ne + e^+ + \nu$

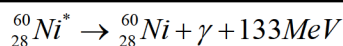


(iii) Gamma Decay

Like an atom, a nucleus also has discrete energy levels - the ground state and excited states. The scale of energy is, however, very different. Atomic energy level spacings are of the order of eV while the difference in nuclear energy levels is of the order of MeV . When a nucleus in an excited state spontaneously decays to its ground state (or to a lower energy state), a photon is emitted with energy equal to the difference in the two energy levels of the nucleus. This is the so-called gamma decay. The energy MeV corresponds to radiation of extremely short wavelength shorter than X ray region. Typically, a gamma ray is emitted when a α or β decay results in a daughter nucleus in an excited state. This then returns to the ground state by a single photon transition or successive transition.

Example of gamma decay





When a beam of γ rays photons passes through matter, the intensity of beam decreases exponentially $I = I_0 \exp(-\mu x)$ where I_0 : Initial Intensity, μ : absorption coefficient of substance, x : thickness of Matter.

Radioactivity

A. H. Becquerel discovered radioactivity in 1896 purely by accident. While studying the fluorescence and phosphorescence of compounds irradiated with visible light, Becquerel observed an interesting phenomenon. After illuminating some pieces of uranium-potassium sulphate with visible light, he wrapped them in black paper and separated the package from a photographic plate by a piece of silver. When, after several hours of exposure, the photographic plate was developed, it showed blackening due to something that must have been emitted by the compound and was able to penetrate both black paper and the silver.

Law of radioactivity

If N is number of nuclei in the sample and ΔN number of nuclei decay in time Δt then

$$\frac{\Delta N}{\Delta t} \propto N \Rightarrow \frac{\Delta N}{\Delta t} = -\lambda N \text{ where } \lambda \text{ is proportionality constant known as decay constant or}$$

disintegration constant, the negative sign arises because number of nuclei in sample will decrease as the time increase. dN is the change in N , which may have either sign. Here it is negative, because out of original N nuclei, ΔN have decayed, leaving $(N - \Delta N)$ nuclei.

$$\frac{\Delta N}{\Delta t} = -\lambda N \text{ as } \Delta t \rightarrow 0 \frac{dN}{dt} = -\lambda N \text{ which is first order differential equation.}$$

$$\int_{N_0}^N \frac{dN}{N} = -\int_{t_0}^t dt \Rightarrow \ln N - \ln N_0 = -(t - t_0) \text{ set } t_0 = 0 \Rightarrow N = N_0 \exp(-\lambda t)$$

Decay Rate (Activity)

The term $R = -\frac{dN}{dt}$ is defined as rate of decay.

$$\frac{dN}{dt} = -\lambda N \Rightarrow -\frac{dN}{dt} = R = \lambda N = \lambda N_0 \exp(-\lambda t)$$

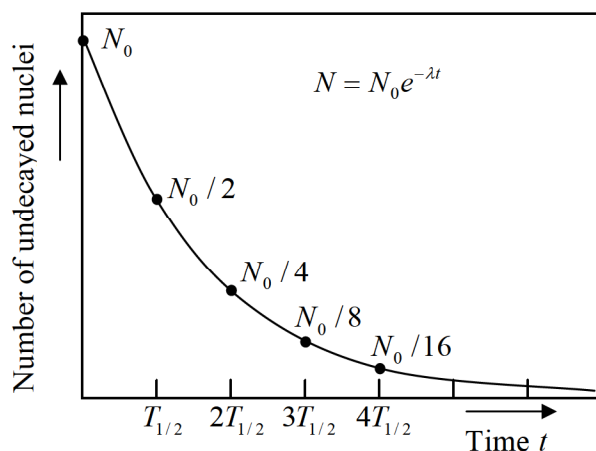


Figure: Exponential decay of a radioactive species. After a lapse of $T_{1/2}$, population of the given species drops by a factor of 2.

where $\lambda N_0 = R_0$ is initial rate or decay rate at $t = 0$ so $R = R_0 \exp(-\lambda t)$

The decay rate of a sample, rather than the number of radioactive nuclei, is a more direct experimentally measurable quantity and is given a specific name: activity. The SI unit for activity is becquerel, named after the discoverer of radioactivity, Henry Becquerel.

Unit of Activity or decay rate $1 \text{ curie} = 1 \text{ ci} = 3.7 \times 10^{10} \text{ decay per sec} = 3.7 \times 10^{10} \text{ Bq}$

Half-life: Half-life of Radionuclide denoted by $t_{1/2}$ is the time it takes for sample that has initially say N_0 radio nuclei to reduce to $\frac{N_0}{2}$ so

$$N = N_0 \exp(-\lambda t) \Rightarrow \frac{N_0}{2} = N_0 \exp(-\lambda t_{1/2}) \Rightarrow t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Clearly if N_0 reduces to half value in time $t_{1/2}$ R_0 will also reduce to $\frac{R_0}{2}$ in time $t_{1/2} = \frac{0.693}{\lambda}$.

Average Mean life: The average life time of radioactive particle before it decay. although they are constant, they are associated with statistical behaviour of population of atom. the life time is given by

$$\tau = \frac{\text{total lifetime of nuclei}}{\text{Total number of nuclei}} = \frac{\int_0^\infty t dN}{N_0} = \frac{\int_0^\infty t \lambda N_0 \exp(-\lambda t) dt}{N_0} = \lambda \int_0^\infty t \exp(-\lambda t) dt \Rightarrow \tau = \frac{1}{\lambda}$$

So $t_{1/2} = 0.693 \cdot \tau$

Radioactivity Equilibrium: If we start with a sample of N_A nuclei with parent nuclide A , after a period of time an equilibrium situation will come about in which each successive daughter $B, C, \dots$ decays at the same rate as it is formed. Thus the activities $R_A, R_B, R_C, \dots$ are equal at equilibrium since $R = \lambda N$ we have radioactive equilibrium $N_A \lambda_A = N_B \lambda_B = N_C \lambda_C = \dots$ where each N_A, N_B and N_C decrease exponentially with time.

Example: The half life of ${}^{238}_{92}\text{U}$ undergoing α decay is 4.5×10^9 years. Find the activity of 1g of sample of ${}^{238}_{92}\text{U}$.

Solution: $t_{1/2} = 4.5 \times 10^9 \text{ Y} = 4 \times 10^9 \times 3.16 \times 10^7 \text{ sec/Y} = 1.42 \times 10^{17} \text{ sec}$

1 gram of uranium consists of $\frac{1}{238 \times 10^{-3}} \text{ kmol} \times 6.025 \times 10^{26} \text{ atom / kmol}$

$$1 \quad R = \lambda N = \frac{0.693}{t_{1/2}} \cdot N = \frac{0.693}{1.42 \times 10^{17}} \times 25 \times 10^{20} \text{ sec}^{-1}$$

$$1.23 \times 10^4 \text{ sec}^{-1} = 1.23 \times 10^4 \text{ Bq}$$

Example: Tritium has half life of 12.5 Y undergoes beta decay. What fraction of sample of pure tritium will remain undecayed after 25 Y

Solution: Let us assume initial atom is N_0 in first 12.5 there will be $\frac{N_0}{2}$ atom will decay so

$$N_0 - \frac{N_0}{2} = \frac{N_0}{2} \text{ will remain.}$$

Now in next 12.5 year half of atom of $\frac{N_0}{2} = \frac{N_0}{4}$ will remain undecayed

Example: The half life of strontium ${}^{90}_{38}\text{Sr}$ is 28.8 Y

(a) Find decay constant

(b) Find initial activity of 1.00g of material

(c) Find the activity after 59.6 Y

Solution: (a) Decay constant is $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{28.8 \times 3.16 \times 10^7 \text{ sec}} = 7.61 \times 10^{-10} \text{ sec}^{-1}$

(b) The atomic mass of ${}^{90}_{38}\text{Sr}$ is 89.91 g Avogadro number is given by $6.022 \times 10^{23} \text{ atom / mol}$

number of atom in 1g of ${}^{90}_{38}\text{Sr}$ is $\frac{1.00 \text{ g}}{89.91 \text{ g}} \times 6.02 \times 10^{23} \text{ nuclei}$

initial activity, $R_0 = \lambda N_0 = (7.61 \times 10^{-10} \text{ sec}^{-1}) \times (6.70 \times 10^{21} \text{ nuclei}) = 5.10 \times 10^{12} \text{ decay/sec}$

$$(c) R = \lambda N(t) = \lambda \times N_0 \times \exp(-\lambda t) = \lambda N_0 \exp\left(-\left(0.693 \frac{t}{t_{1/2}}\right)\right)$$

$$t = 59.6Y \quad t_{1/2} = 28.8Y$$

$$R(t) = \lambda N_0 \exp\left(-\left(0.693 \frac{t}{t_{1/2}}\right)\right) = R_0 \exp(-0.693 \times 2) = R_0 \times \exp(-1.386) = 0.25 R_0$$

$$= 0.25 \times 5.10 \times 10^{12} \text{ decay/sec} = 1.275 \text{ decay/sec}$$

Example: Approximately 20% of human body by mass is carbon. Calculate the activity due to $^{14}_6\text{C}$ in 1 kg of carbon found in living organism. Express the activity in the unit of *Bq* and *Ci*.

Solution: The abundance of ^{14}C in carbon sample is 1.3×10^{-12} . half-life of ^{14}C is 5730 year.

Number of carbon ^{12}C nuclei in 1kg of sample

$$N(^{12}\text{C}) = \frac{6.02 \times 10^{23} \text{ mol}^{-1}}{12.0 \text{ g/mol}} \times 1000 \text{ g} = 5.02 \times 10^{25}$$

$$\text{Number of carbon } N(^{14}\text{C}) = (5.02 \times 10^{25}) (1.3 \times 10^{-12}) = 6.52 \times 10^{12}$$

Now we can find the activity *A* by using the equation

$$A = \lambda N = \frac{0.693}{t_{1/2}} N = \frac{0.693}{5730} \times 6.52 \times 10^{12} \text{ y}^{-1} = \frac{7.89 \times 10^9 \text{ y}}{3.16 \times 10^7 \text{ s}} = 250 \text{ Bq}$$

$$A = \frac{250}{3.7 \times 10^{10} \text{ Bq/Ci}} = 6.76 \times 10^{-9} \text{ Ci} \Rightarrow 6.76 \text{ nCi}$$

Example: In an ancient burial cave, a team of archaeologist discovers ancient wood furniture.

Only 80% of the original ^{14}C remains in wood. How old is furniture? Half life of ^{14}C is 5730 year.

Solution: $\frac{N}{N_0} = \exp(-\lambda t)$ it is given $\frac{N}{N_0} = 0.80 \Rightarrow \exp(-\lambda t) \Rightarrow -\lambda t = \ln 0.80 = -0.223 = -\lambda t$

$$\text{So } t = \frac{0.223}{\lambda} \Rightarrow \frac{0.223}{\frac{0.693}{t_{1/2}}} = \frac{0.223}{0.693} \times 5730 \text{ y} = 1844 \text{ y}$$

Example: The atomic ratio between the uranium isotopes ^{238}U and ^{234}U in a mineral sample is found to be 1.8×10^4 . The half-life of ^{234}U is 2.5×10^5 y. Find the half-life of ^{238}U .

Solution: $N(238) \times \lambda(238) = N(234) \lambda(234) \Rightarrow N(238) \times \frac{0.693}{t_{1/2}(238)} = N(234) \times \frac{0.693}{t_{1/2}(234)}$

$$= t_{1/2}(238) = \frac{N(238)}{N(234)} \times t_{1/2}(234) \Rightarrow t_{1/2}(238) = (1.8 \times 10^4)(2.5 \times 10^5 \text{ y}) = 4.5 \times 10^9 \text{ y}$$

Example: A substance X disintegrates into substance Y find the number of nuclei of Y which is expected number of nuclei N_Y still undecayed. Where N_{X_0} and N_{Y_0} are initial amount of X and Y respectively. λ_x and λ_y are decay constant for X and Y .

Solution: $\frac{dN_x}{dt} = -\lambda_x N_x \Rightarrow \frac{dN_x}{dt} = -\lambda_x N_{X_0} \exp(-\lambda_x t)$

For X nucleus disintegrate with rate of $\lambda_x N_x$ which is also rate at which Y is formed .so decay

equation for Y is given by $\frac{dN_Y}{dt} = \lambda_x N_x - \lambda_y N_Y \Rightarrow \frac{dN_Y}{dt} = \lambda_x N_{X_0} \exp(-\lambda_x t) - \lambda_y N_Y$

$$N_Y \lambda_y + \frac{dN_Y}{dt} = N_{X_0} \lambda_x \exp(-\lambda_x t) \Rightarrow N_Y \lambda_y dt + dN_Y = N_{X_0} \lambda_x \exp(-\lambda_x t) dt$$

Multiply both side by $\exp(\lambda_y t)$

$$\exp(\lambda_y t) N_Y \lambda_y dt + \exp(\lambda_y t) dN_Y = N_{X_0} \lambda_x (\exp \lambda_y t) \exp(-\lambda_x t) dt$$

$$d(N_Y \exp(\lambda_y t)) = N_{X_0} \lambda_x (\exp \lambda_y t) \exp(-\lambda_x t) dt$$

$$N_Y \exp(\lambda_y t) = \frac{N_{X_0} \lambda_x}{\lambda_y - \lambda_x} \exp(\lambda_y t - \lambda_x t) + C \text{ at } t = 0, N_Y = N_{Y_0} \Rightarrow C = N_{Y_0} - \frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x}$$

$$N_Y = \left(N_{Y_0} - \frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x} \right) \exp(-\lambda_y t) + \left(\frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x} \right) \exp(-\lambda_x t)$$

Nuclear Reaction: Nuclear reaction is a process in which two nuclei, or a nucleus and an external subatomic particle, collide to produce one or more new nuclides. Thus, a nuclear reaction must cause a transformation of at least one nuclide to another. If a nucleus interacts with another nucleus or particle and they then separate without changing the nature of any nuclide, the process is simply referred to as a type of nuclear scattering, rather than a nuclear reaction.

Typically, nuclear reaction represented in term of equation as $x + X = Y + y$ where x projectile hits the target X and Y and y are out going particle.

The Q value is defined as total change in kinetic energy.

Total energy of particle X is given by $E_X = T_X + M_X c^2$ where T_X is kinetic energy and $M_X c^2$ is rest mass energy

Total energy of particle x is given by $E_x = T_x + M_x c^2$ where T_x is kinetic energy and $M_x c^2$ is rest mass energy

Total energy of particle Y is given by $E_Y = T_Y + M_Y c^2$ where T_Y is kinetic energy and $M_Y c^2$ is rest mass energy

Total energy of particle y is given by $E_y = T_y + M_y c^2$ where T_y is kinetic energy and $M_y c^2$ is rest mass energy

From the conservation energy $E_X + E_x = E_Y + E_y$

$$(T_X + M_X c^2) + (T_x + M_x c^2) = (T_Y + M_Y c^2) + (T_y + M_y c^2)$$

The Q value is given by

$$(T_Y + T_y) - (T_X + T_x) = [(M_X + m_x) - (M_Y + M_y)] c^2$$

So
$$Q = [(M_X + m_x) - (M_Y + M_y)] c^2$$

This equation will mainly valid for one dimensional collision.

Some other facts about Q value

(1) The KE_α of the emitted α -particle is never quite equal to Q , since momentum must be conserved, the nucleus recoils with a small amount of kinetic energy when the α -particle emerges. Thus $KE_\alpha \approx \frac{A-4}{A} \cdot Q$

(2) Center of mass coordinate system

Most of nuclear reaction in a laboratory occur when a moving nucleons or nucleus strikes a stationary one. analysing such a reaction is simplified when we use a coordinate system that moves with the center of mass of colliding particles.

The observer located at center of mass, the particle has equal and opposite momenta .Hence particle of mass m_A with speed v approaches to stationary particle of mass m_B . As viewed from laboratory ,the speed v_{cm} is speed of center of mass

$$m_A(v - v_{cm}) = m_B v_{cm} \Rightarrow v_{cm} = \left(\frac{m_A}{m_A + m_B} \right) v$$

In most of the nuclear reaction takes place $v \ll c$ so nonrelativistic treatment is sufficient

Kinetic energy in lab frame is given by $T_{lab} = \frac{1}{2} m_A v^2$

Kinetic energy in center mass frame is given by $T_{cm} = \frac{1}{2} m_A (v - v_{cm})^2 + \frac{1}{2} m_B v_{cm}^2$

$$= \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v_{cm}^2 - \frac{1}{2} 2 m_A v v_{cm} + \frac{1}{2} m_B v_{cm}^2$$

$$= \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v_{cm}^2 - \frac{1}{2} 2 m_A \left(\frac{m_A + m_B}{m_A} \right) v_{cm} v_{cm} + \frac{1}{2} m_B v_{cm}^2$$

$$\frac{1}{2} m_A v^2 - \frac{1}{2} (m_A + m_B) v_{cm}^2 = \frac{1}{2} m_A v^2 - \frac{1}{2} (m_A + m_B) \left(\frac{m_A}{m_A + m_B} \right)^2 v^2$$

$$= \frac{1}{2} m_A v^2 - \frac{1}{2} \left(\frac{m_A}{m_A + m_B} \right) m_A v^2$$

$$T_{cm} = T_{lab} - \frac{m_A}{m_A + m_B} T_{lab} \Rightarrow T_{cm} = \left(\frac{m_B}{m_A + m_B} \right) T_{lab}$$

Here we can see that $T_{cm} < T_{lab}$

The Q value of nuclear reaction $A + B \rightarrow C + D$ is given by $Q = (m_A + m_B - m_C - m_D) c^2$

If $Q > 0$ K.E. of product is **more than** kinetic energy of reactant, which means the reaction is exothermic reaction. The energy is given off by the reaction.

If $Q < 0$ K.E. of product is **less than** kinetic energy of reactant, which means the reaction is endothermic reaction. The energy will flow into system.

Which means enough kinetic energy KE_{cm} in the center of mass frame must be provided by the reacting particle so that $KE_{cm} + Q \geq 0$

Example: The polonium isotope $^{210}_{84}\text{Po}$ is unstable and emits a 5.30MeV alpha particle. The atomic mass of $^{210}_{84}\text{Po}$ is $209.989u$ and that of ^4_2He is $4.006u$. Identify the daughter nuclei and find its mass.

Solution: $^{210}_{84}\text{Po} = ^{206}_{82}\text{Po} + ^4_2\text{He}$ so new particle is $^{206}_{82}\text{Po}$ let us assume its mass is m_f .

The kinetic energy of alpha particle is $KE = 5.34\text{MeV}$

So, we can calculate the Q Value. $Q = \frac{A}{A-4} K.E_\alpha = \left(\frac{210}{210-4} \right) 5.30 \text{ MeV} = 5.40 \text{ MeV}$

The equivalent mass is given by $\frac{5.40 \text{ MeV}}{931 \text{ MeV/u}} = 0.0058 \text{ u}$

So $m_Q = m_i - m_\alpha - m_f \Rightarrow m_f = m_i - m_\alpha - m_Q$

$$209.9829 \text{ u} - 4.0026 \text{ u} - 0.0058 \text{ u} = 205.9745 \text{ u}$$

Example: Find the minimum kinetic energy in laboratory frame needed by an alpha particle to cause the reaction $^{14}\text{N}(\alpha, p)^{17}\text{O}$. The masses of ^{14}N , ^4He , ^1H and ^{17}O are respectively 14.00307 u , 4.00260 u , 1.00783 u and 16.99913 u

Solution: The reaction is $^{14}\text{N}(\alpha, p)^{17}\text{O}$ which means $^4_2\text{He} + ^{14}_7\text{N} \rightarrow ^{17}_8\text{O} + p$

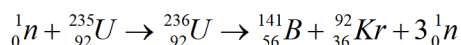
$Q = (14.00307 \text{ u} + 4.00260 \text{ u} - 1.00783 \text{ u} - 16.99913 \text{ u}) \times 931 \text{ MeV/u} = -1.20 \text{ MeV}$ which is negative so reaction is endothermic in nature.

The minimum kinetic energy in centre of mass frame is $T_{cm} = 1.20 \text{ MeV}$ in order to reaction

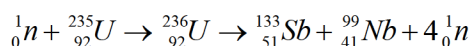
occur. $T_{lab} = \left(\frac{m_A + m_B}{m_B} \right) T_{cm} = \left(\frac{4.00260 + 14.00307}{14.00307} \right) 1.20 \text{ MeV} = 1.54 \text{ MeV}$

Nuclear fission: Nuclear fission is a nuclear reaction in which the nucleus of an atom breaks into two or more smaller nuclei. The fission process often produces gamma photon, and releases a very large amount of energy even by the energetic standards of radioactive.

Some important nuclear fission



The same reaction can produce other pair of intermediate mass fragments



Another example $^1_0n + ^{235}_{92}\text{U} \rightarrow ^{140}_{54}\text{Xe} + ^{94}_{38}\text{Sr} + 2^1_0n$

Example: For the reaction $^1_0n + ^{235}_{92}\text{U} \rightarrow ^{236}_{92}\text{U} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3^1_0n$, find how much energy released in the process.

Solution: Mass of $^{235}_{92}\text{U} = 235.045733 \text{ u}$, Mass of $^1_0n = 1.008665 \text{ u}$

Total initial mass = 236.054398 u .

Mass of $^{141}_{56}\text{Ba} = 140.99177 \text{ u}$, Mass of $^{92}_{36}\text{Kr} = 91.8854 \text{ u}$, Mass of $3^1_0n = 3.0259 \text{ u}$,

Total final mass = 235.00307 u .

Mass defect (Δm) = initial mass - final mass = 1.051328 u .

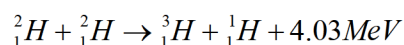
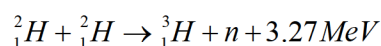
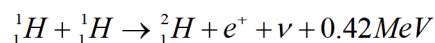
Energy released i.e. Q -value = $\Delta m \times 979.312032 \text{ MeV} \approx 200 \text{ MeV}$.

Number of uranium nucleus in 1 kg of uranium = $6.023 \times 10^{23} \times 1000/235$.

Energy released by 1 kg of uranium = $6.023 \times 10^{23} \times 1000/235 \times 200 \text{ MeV} = 5.13 \times 10^{26} \text{ MeV}$

Nuclear fusion is a nuclear reaction in which two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particles (neutrons or protons). The difference in mass between the reactants and products is manifested as either the release or the absorption of energy.

Some important nuclear fusion reaction



Nuclear Fusion in Stars

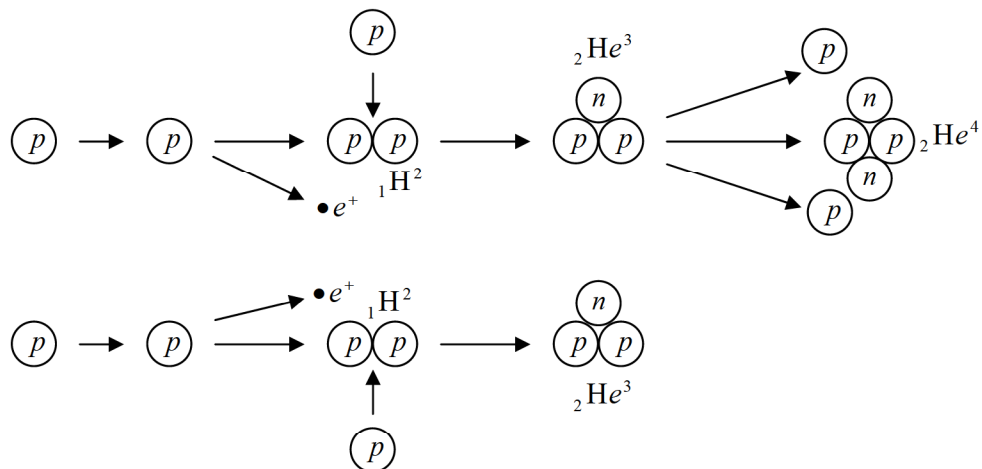
Fusion is a process, in which the lighter nuclei fuse together and produce a heavier nucleus. The sum of the masses of the individual light nuclei is more than would be the mass of the nucleus formed by their fusion, and thus the fusion process result in liberation of energy.

Self-sustaining fusion reactions can occur only under conditions of **extreme temperature** and **density**. The high temperature ensures that the same nuclei have the energy needed to come close enough together to interact, and the high density ensures that such collisions are frequent. A further condition for the proton-proton cycles is a large reacting mass, such as that of the sun, since much time may elapse between the initial fusion of a particular proton and its eventual incorporation in an α -particle.

The basic energy producing process in the sun is the fusion of hydrogen nuclei into helium nuclei. This can take place in several different reaction sequences, the most common of which, the **proton–proton cycle**. The total evolved energy is 24.7 MeV per ${}_2^4\text{He}$ nucleus formed.

Since 24.7 MeV is $4 \times 10^{-12} \text{ J}$, the sun's power output of $4 \times 10^{26} \text{ W}$ means each sequence of reactions must occur 10^{38} times per second.

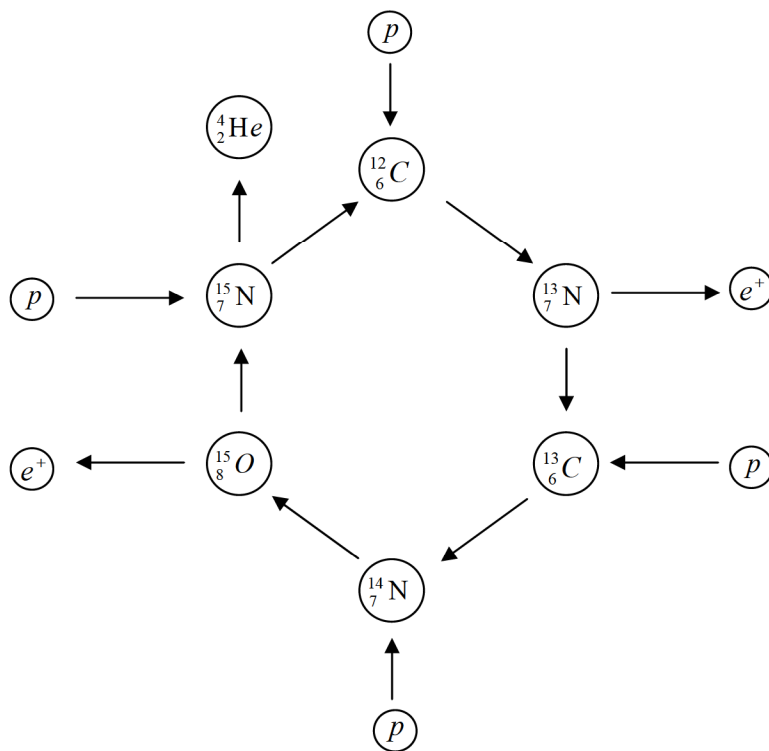
- Proton-Proton cycle**



Energy is given off at each step. The net result is the combination of four hydrogen nuclei to form a helium nucleus and two positrons.

- Carbon-Cycle**

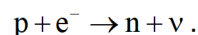
It also involves the combination of four hydrogen nuclei (Stars hotter than sun) to form a helium nucleus with the evolution of energy. The net result again is the formation of an α -particle and two positrons from four protons, with the evolution of 24.7 MeV. The initial ${}^{12}_6\text{C}$ acts as a kind of catalyst for the process, since it reappears at its end.



Some other nuclear reaction

Electron capture

It is closely connected with positron emission. In electron capture a nucleus absorbs one of its inner atomic electrons, with the result that a nuclear proton becomes a neutron and neutrino is emitted:



Usually, the absorbed electrons come from the *K*-shell, and an X-ray photon is emitted when one of the atoms outer electrons falls into the resulting vacant state. The wavelength of the photon will be one of those characteristics of daughter element, not of the original one, and the process can be recognized on that basis.

Note: 1. Electron capture is competitive with positron emission since both processes lead to the same nuclear transformation.

2. Electron capture occurs more often than positron emission in heavy nuclides because the electrons in such nuclides are relatively close to the nucleus, which promotes their interaction with it.

Inverse Beta Decay: In this reaction proton will absorb antineutrino and produce neutron and positron or neutron will absorb neutrino and produce electron. they can be shown as reaction respectively $p + \bar{\nu} = n + e^+$ and $n + \nu = p + e^-$

These reactions are very low probability which is why neutrinos and antineutrino are able to pass vast amount of matter.